

High-Density Cable Employing 160 μm Reduced-Cladding 4-Core Multi-Core Fibre for Data Centre Applications

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Abstract A high-density 288-fibre cable was fabricated employing reduced-cladding 100/160 μm , 4-core, multi-core fibres (MCFs) for intra-/inter-data centre links (lengths <10 km). The cable suppressed the microbending loss and maintained practical crosstalk and temperature stability levels, demonstrating the feasibility of reduced-cladding MCF cabling for practical deployment. ©2026 The Author(s)

Introduction

The computational requirements of artificial intelligence (AI) training and inference are increasing. Therefore, high-density graphics processing unit (GPU) implementations are being pursued in AI data centres. Given the increasing density of GPUs, interconnects between GPUs with higher density and transmission capacity are required. Multi-core fibres (MCFs) are a promising solution for realising high-density optical interconnections. Furthermore, the industry standardisation of MCF-based optical interconnects for practical deployment has recently progressed. In 2026, a Multi-Source Agreement on an MCF design for O-band optical communications in AI data centre campuses was announced [1]. In the International Telecommunication Union Telecommunication (ITU-T) standardisation sector, discussions are underway regarding the standardisation of weakly coupled MCFs with a standard cladding diameter, with a focus on 2-core (2c) and 4-core (4c) MCFs [2]. MCFs with core densities higher than 4c-MCFs have been proposed. Higher core densities can be achieved by restricting the operation to the O-band and reducing the coating thickness. Because the O-band transmission is primarily used in data centres, restricting the transmission band is feasible for this application. Even when these two approaches are adopted, the choice of cladding diameter remains flexible.

An 8-core (8c)-MCF has been reported as an MCF with a high core density and a standard cladding diameter [3]. MCFs with a standard cladding diameter have the advantage of being compatible with existing systems. A 16-core (16c)-MCF with a cladding diameter of 190 μm and a coating diameter of 250 μm (190/250 μm) has been reported as an MCF with a larger cladding diameter [4]. This type of fibres offer a higher core density and improved resistance to microbending

loss. A 4c-MCF with a 100/160 μm structure has also been reported as an MCF with a reduced cladding diameter [5]. For reduced-cladding MCFs, their smaller fibre cross-sectional areas allow a longer fibre length to be obtained from a given preform volume.

Because each fibre design offers different advantages, the appropriate design should be selected according to the target application. In addition, cables using high-core-density MCFs have been reported for practical applications. The cabling of 8c-MCFs has been reported for fibres with standard cladding diameters [6]. For fibres with larger cladding diameters, cabling of a 12-core (12c)-MCF has also been reported [7]. However, to the best of our knowledge, there are no reports on the cabling of MCFs with reduced cladding diameters. For reduced-cladding fibres, the increased microbending loss owing to the reduced coating and cladding diameters is a major concern.

In this study, we fabricated a high-density cable employing a 100/160 μm 4c-MCF. The 100/160 μm 4c-MCF offers a higher core density than a 125/250 μm 8c-MCF and retains the square 4c arrangement for which related technologies have been most extensively studied. Temperature-dependent tests were also conducted to assess the effect on the microbending loss.

Design and Characteristics of the Fibre

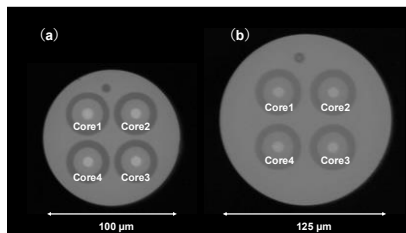
Microbending loss can be evaluated using an analytical model for dual-coated optical fibres [8]. According to this model, reducing the coating thickness and cladding diameter increases the microbending loss. Therefore, thin coated fibres with reduced cladding diameters are more susceptible to microbending than those with standard cladding diameters. Fibres with reduced cladding must be designed such that they can withstand cabling. Sillard *et al.* reported that trench-

Tab. 1: Optical characteristics of the fabricated MCF (before coloured).

Attribute	Cladding diameter	Coating diameter	Attenuation	MFD	MBL	Cutoff wavelength	Total XT
Unit	μm	μm	dB/km	μm	dB/turn at R=10	nm	dB at 1 km
Wavelength	-	-	1310 nm	1310 nm	1550 nm	n/a	1310 nm
C1	99.0	155.1	0.37	8.6	0.01	1,173	-55
C2			0.38	8.4	0.01	1,196	-56
C3			0.38	8.4	0.03	1,172	-59
C4			0.38	8.4	0.06	1,180	-58
ITU-T G.657.A2	-	-	< 0.4	8.6-9.2 $\pm$ 0.4	< 0.1	< 1260	-

assisted (TA) profiles provided higher resistance to microbending than step-index (SI) profiles [9]. Therefore, a TA profile was adopted for the core design. In addition, the thicknesses of the primary and secondary coatings were optimised using an analytical model for microbending loss in dual-coated optical fibres [8]. This fibre is intended for O-band inter- and intra-data centre connectivity for distances up to 10 km. To accommodate four cores within a 100 μm cladding, both crosstalk (XT) and confinement loss must be suppressed. In this respect, the TA profile was more advantageous than the SI profile. When a mass-producible core design with a trench depth of -0.3% is used, the optical characteristics required for the target application can be achieved with a core pitch of 35 μm [5]. Therefore, the core pitch was set to 35 μm .

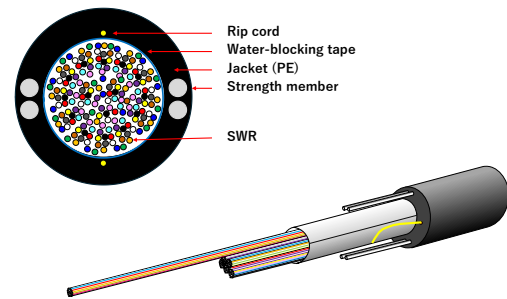
Based on this design, 25.7 and 18.8 km 4c-MCFs were fabricated. Fig. 1 shows cross-sectional images of the fabricated fibre and a 125 μm cladding fibre for size comparison. The average optical characteristics are summarised in Tab. 1. The key optical properties satisfied the ITU-T G.657.A2 standard. The maximum total XT at 1310 nm was -55 dB at 1 km.

**Fig. 1:** Cross-sections of (a) the fabricated fibre and (b) a standard cladding fibre.

Fabricated Cable With Ribbonised Fibre

We fabricated a 288-fibre cable with a length of 1.2 km and an outer diameter of 10.5 mm. The cable consisted of 24 ribbons, each comprising 12 optical fibres arranged in an intermittently bonded spider web ribbon (SWR[®]) structure. Among the 24 ribbons, 12 contained 100/160 μm 4c-MCFs. In each of these ribbons, one fibre was a 4c-MCF, and the remaining 11 fibres were

dummy fibres. The intermittently bonded structure provided high flexibility and was suitable for high-density packaging. A high-density cable was achieved by assembling the ribbons into a Wrapping Tube Cable (WTC[®]) structure, in which they were wrapped with water-blocking tape and covered with a polyethylene jacket containing two pairs of strength members. The ribbon and cable are shown in Fig. 2.

**Fig. 2:** Cross-section of the fabricated cable.

Tab. 2 compares the cable diameter and weight of 288-fibre SWR/WTC cables fabricated using 250 and 160 μm fibres. If all 288 fibres in the 160 μm fibre cable are 4c-MCFs, the core count per unit cable cross-sectional area is 5.2 times higher, and the cable weight is 33% lower than that of a cable using 250 μm , single-core fibres (SCFs).

Tab. 2: Dimensions of the fabricated 288-fibre WTCs

	Unit	250 μm fibre	160 μm fibre
Outer diameter	mm	12.0	10.5
Weight	kg/km	105	70

Fig. 3 shows the attenuation distributions at 1310 nm for the bare fibre, ribbon, and cable conditions. For the bare fibre condition, the data were obtained from the two MCFs, whereas for the ribbon and cable conditions, the data were obtained from the twelve MCFs measured in each state. In each case, the vertical axis indicates the fraction of the total fibres. Although some fibres in the ribbon and cable conditions exhibited higher attenuation than those in the bare fibre condition, the average attenuation of the ribbon and cable was lower than that of the bare fibre condition. The

increase in attenuation observed in some fibres was attributed to the fact that this was the first ribbon fabrication trial, and this increase could be reduced by optimising the fabrication conditions. The improvement in the average attenuation relative to the bare-fibre condition was mainly attributed to the difference in the spool diameter.

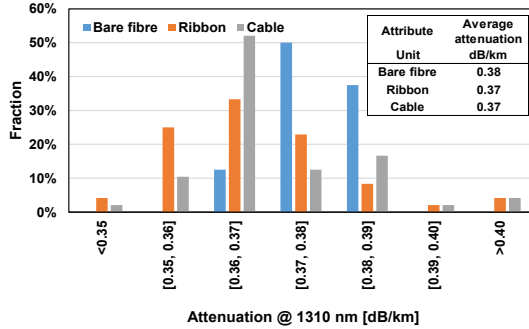


Fig. 3: Attenuation of all tested conditions.

Temperature-Dependent Attenuation and XT Variation

The fabricated cable was subjected to a temperature cycling test over two cycles between -40 and 70 °C. The effects of temperature variation on the fan-in/fan-out devices were eliminated by fusion-splicing them to both ends of the cable outside the temperature chamber. Fig. 4 shows the attenuation increase at each temperature relative to the initial condition at 20 °C. The points indicate the average values, and the error bars represent the maximum and minimum values. The maximum increase in attenuation caused by the temperature change was 0.011 dB/km. Sasaki *et al.* reported that in a cable employing $125/200$ μm 4c-MCFs, the attenuation increase during temperature cycling tests at 1550 and 1310 nm was less than 0.05 dB/km, demonstrating characteristics comparable to those of SCFs [10]. The increase in the attenuation observed in this study was smaller than that reported previously. These results indicate that the excess attenuation caused by the temperature change was effectively suppressed.

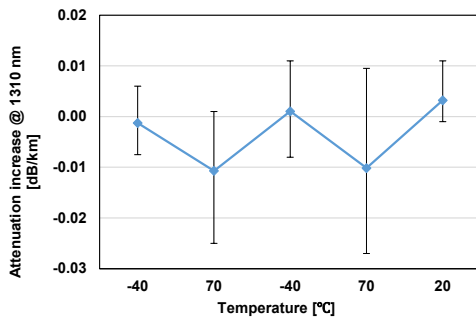


Fig. 4: Temperature cycling test results.

The effect of temperature on the XT was also investigated. The XT was measured at -40 , 20 , and 70 °C, and the results are shown in Fig. 5.

Even in the worst case, the total XT was -45.3 dB at 1 km. A core-to-core XT of ≤ -34.2 dB at 1 km, corresponding to an aggregate XT of ≤ -40 dB at 10 km in bi-directional transmission, is sufficiently low for IM-DD PAM4 transmission [7]. Therefore, the present cable is feasible for practical applications.

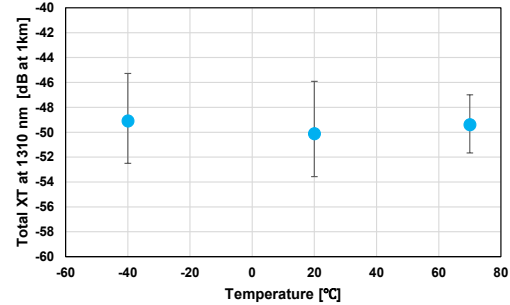


Fig. 5: Temperature dependence of XT.

Herein, we discuss the reasons for the favourable results obtained in the temperature tests. Cable shrinkage at low temperatures was considered to induce macro and microbending of the fibre, which may have caused variations in the attenuation and XT. No excess loss was observed with respect to attenuation. This was attributed to the TA profile, which is effective against both macro and microbending, and the optimised coating thickness, which improves the resistance to microbending. No clear temperature dependence was observed for XT. In homogeneous MCFs, bending can change the propagation constant mismatch between the cores [11], thereby suppressing inter-core coupling and reducing XT. However, in the proposed fibre, the TA profile likely mitigated the bending effect itself, therefore, the propagation constant mismatch induced at low temperatures was small, and no clear reduction in XT was observed. In addition, microbending only has a minor effect on the XT among homogeneous cores [12]. These results indicate that the present fibre structure is highly resistant to bending caused by low-temperature shrinkage, with minimal degradation in the attenuation or XT.

Conclusion

A 288-fibre, high-density cable employing $100/160$ μm 4c-MCFs was developed. The most critical issue in cables using reduced-cladding fibres is the potential increase in attenuation caused by microbending loss. By optimising the TA profile and coating diameter, the excessive increase in loss was suppressed. Compared with a cable that used 250 μm SCFs, the number of cores per unit cable cross-sectional area was increased by a factor of 5.2 , whereas the cable weight was reduced by 33% . The proposed cable is a promising solution for data centres requiring high-density optical wiring.

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